

COLE RICHARDS

Statistics & Data Science | Environmental Data Science | GIS | Applied Modeling | Operations Software

Los Angeles / Santa Monica, CA

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SUMMARY

UCLA Statistics & Data Science student graduating June 2026. Experience in environmental modeling, geospatial analysis, and operational software—spanning startup operations, GIS-based wildfire risk targeting, internal tool development, and field execution. Interested in roles where statistical methods and spatial data drive real-world decisions.

EDUCATION

University of California, Los Angeles

Sep 2024 – Jun 2026

B.S. Statistics & Data Science

Focus: statistics, GIS, geospatial analysis, applied modeling

Santa Barbara City College

Aug 2022 – May 2024

A.A. Mathematics | A.A. Computer Science

GPA: 3.98 | *President's Honor Roll*

TECHNICAL SKILLS

Languages: Python, R, SQL, JavaScript, HTML/CSS, Swift

Libraries: pandas, NumPy, tidyverse, ggplot2, geoR, gstat

Tools: ArcGIS, QGIS, Supabase, Mapbox, Vite, Netlify, Excel

Areas: Geospatial analysis, statistical modeling, kriging, regression, GIS, forecasting, workflow automation, dashboard design

EXPERIENCE

Co-Founder & COO | Trident Ember Defense

Aug 2025 – Present

- Built and deployed 6 internal and external tools—website, customer-facing planner/quoter, CRM, 3D print inventory queue, 2D mapping/inventory tool, and quote builder
- Designed dozens of exterior wildfire sprinkler systems, completing 3 installations and generating \$20K+ in revenue
- Performed geospatial wildfire risk analysis using ArcGIS and QGIS to drive client targeting and project planning
- Led field installation execution, managing an 8-person crew on a 3,400 sq ft residential project

Inventory Specialist | UCLA Housing IT

Aug 2024 – Dec 2025

- Managed thousands of technology assets and hundreds of toner units across 12+ product types in campus storage
- Built an inventory logging system to replace a static sheet, enabling change history tracking and usage trend analysis
- Supported device setup, intake processing, and systems accuracy for campus housing operations

Accounts Analysis & Development | Esperer Holdings

Oct 2020 – Oct 2021

- Automated Excel-based invoice tracking and reporting workflows across thousands of staffing client line items
- Reduced a week-long manual analysis and reporting cycle to minutes of automated input, freeing 35+ hours per week
- Trained 3 team members on the new automated process to ensure continuity

PROJECTS

Wildfire Risk Spatial Analysis | QGIS, ArcGIS, R, NDVI

Multi-layer GIS workflow combining WUI zones, Fire Hazard Severity Zones, historical fire perimeters, and NDVI vegetation data to identify at-risk properties across California for targeted outreach.

Geostatistical Kriging Analysis | R, geoR, gstat

Applied ordinary and universal kriging to spatial datasets using variogram fitting and PRESS cross-validation to model and predict environmental conditions across unsampled locations.

Airbnb Amenity Pricing Analysis | R, tidyverse, ggplot2

Built LASSO logistic regression models with phrase-level amenity tokenization to identify which amenities predict above-median pricing, evaluated via AUC with neighborhood-stratified confounding analysis.

Operations & Planning Tools | Supabase, Mapbox, JS, Vite, Netlify

Full-stack internal system connecting CRM, quoting, inventory, and sprinkler layout planning into a unified workflow for Trident Ember Defense.

Interests: Surfing, volleyball coaching, hiking, yoga & meditation